Problem Set 3
ECO3121 - Fall 2022

November 8, 2022

Due 5 PM, 11/18/2022

Download *aghousehold.dta* and *village_rainfall.dta* datasets from the blackboard site and load into STATA. The main data we use is the National Fixed Point Survey (NPFS), which is a nationally representative panel dataset (unbalanced) of roughly 5000 households in 88 villages between 1995 and 2002. It is collected by the Ministry of Agriculture and Rural Affairs of China.

On the blackboard, we also uploaded a second dataset regarding precipitation records in each village in 2001. See the variable list below. Write up the answers to 1) - 9) below. In addition, also attach the do file that you used to answer the following questions.

variable name	type	format	label
vl_id	int	%8.0g	village ID, consistent with household data
lat_o	float	%8.0g	latitude of each village
lon_o	float	%8.0g	longitude of each village
provname	str24	%24s	province name in Chinese
cityname	str33	%33s	city name in Chinese
countyname	str33	%33s	county name in Chinese
year	float	%9.0g	year of the record
av_rain	float	%9.0g	average precipitation in 2001, measured by mm
sd_rain	float	%9.0g	standard deviation of precipitation across months in 2001
z_rain	float	%9.0g	zero score of precipitation in 2001

Question 1

First, limit the sample to households **in year 2001** by running code "keep if year==2001" in Stata. you can generate *variable* yield (output per unit of land) via $\frac{hx95_nh125}{hx95_nh112}$, and variable *fertilizer* (fertilizer application per unit of land) through $\frac{nh6_3}{hx95_nh112}$.

1. The variable *h95_nh269* indicates how many days for household members in each household had been working as temporal migrants (measured by days) in 2001.

   Generate a binary variable regarding household migration decision. It takes the value of 1 if *h95_nh269* is greater than zero, otherwise it is 0. Generate the natural

log of yield and fertilizer application intensity (using log(fertilizer + 1) and log(yield + 1) to smooth zero values.).

Export a table that lists the number of observations, the mean, the standard deviation for the above three variables. Briefly describe what we learn from the table about the sample data. (3 points)

2. Run a linear probability model of household migration decision in question (1) on the natural log of yield.
   Interpret the result and comment on its statistical significance. (4 points)

3. Run a probit model of household migration decision in question (1) on the natural log of yield.
   Interpret the results if the value of *logyield* increases from 5 to 8, and comment on its statistical significance. (4 points)

4. List three plausible arguments why the point estimate in question (2) and (3) could be biased, and the corresponding bias directions (**upwards** or **downwards**) relative to the true causal effect of household's agricultural production on household migration decision. (6 points)

5. Now your friend suggests that the fertilizer application intensity of each household could be a valid instrumental variable (IV) for your measure of household's yield. List the assumptions that need to hold true for this to be correct and explain why this instrumental variable is true (or not true). (4 points)

6. Now your TA suggests that the rainfall (precipitation) could be a valid instrumental variable (IV) for your measure of household's yield. Try to merge the household's production dataset and precipitation dataset via *vl_id*, the specific village identifier, using stata command *merge* (Many-to-one merge, type *"help merge"* in Stata for assistance).

   You decide to use the average rainfall in 2001 (*av_rain*) as the IV for the household's yield. Verify if the assumption of instrument relevance is satisfied using the first stage regression, and export the results. **Write down the first stage regression model** and interpret the result and statistical significance of your result. (5 points)

7. Now use Stata/R to estimate the 2nd stage IV point estimate (using linear probability model) as suggested by the TA, and export your result. **Write down the second stage regression model** and interpret the result and statistical significance of your result. (5 points)

8. Now your TA tells you that you can use *ivreg* command directly to replicate the results in question (7). Do you find any difference in the IV estimations (β_{IV}) **regarding the coefficients and standard errors** relative to (7). (3 points)

In reference to your answer to question (2), is the difference between the linear probability model and IV point estimates as you expected or rather not? (2 points)

9. Now your professor suggests that the precipitation in each village is likely correlated with other local characteristics that affect household's migration decision, such as the logarithm of the household consumption (log(nh7_40 + 1)), the logarithm of the household income (log(nh7_1 + 1)). Propose a way to verify whether these concerns are relevant and export your regression results in the same table as before. Comment on the results and what they imply about the validity of the instrumental variable strategy. (4 points)